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Gantry Display ESD Rework

christopher hardiman (christopher c hardiman), 09/26/2002 - 10:39 am

**Priority**

4 - Tips, Parts, Surveys, etc.

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Communique Number

MQUE-5ECKYF

Equipment

LightSpeed Plus & Ultra

Problem

Recently, a retest of ESD immunity on a LightSpeed Ultra System was performed at CT Engineering. Due to a procedural change, the testing failed three (3) Ultra/Plus gantry displays in a row. The new procedure included wiping the non-conductive front plastic of the display with a damp cloth between sweeps of an ESD gun. This added moisture caused the plastic cover to draw an arc. The current discharged behind the gantry cover at the metal faceplate of the display. Previously, failures of this type were sporadic and difficult to replicate. At issue was the discovery of some "floating metal" that was missed in the design. This has been identified as a display problem only, and not a gantry pushbutton issue. Pushbuttons have no floating metal and pass ESD testing; however, a problem of this nature could create havoc on the entire CAN network for the controls - especially in dry environments.

Solution

The addition of a grounding location for the display board has been proven to eliminate ESD failures induced in dry environments from the front of the gantry display. After the addition of the ground, a full 25,000 Volt charge was safely dissipated by three (3) different displays in three (3) successive tests. See below document for details.

You can download any of the files below by dragging them to your Windows desktop. To open a file for reading, double-click it.

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Revision

09/26/2002 10:39 AM

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GE Medical Systems

Technical Publication

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Revision 0**

GE Medical Systems LightSpeed Plus & Ultra Gantry Display ESD Rework

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Revision History

REV	DATE	REASON FOR CHANGE
Rev 0	9/26/2002	Initial Release

List of effective pages

Page	Rev.	Page	Rev.	Page	Rev.
Title Page	0	1	0	Rear	-
Rev/LOEP	0				

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Materials:

- Small piece of copper tape
- Solder

Tools:

- Soldering iron
- Flat blade screw driver
- #1 Phillips screw driver
- Small adjustable wrench

Procedure:

Where necessary, please refer to your System Service Manual in the performance of these tasks.

1. Remove gantry side and top covers.
2. Disable Axial Drive.
3. Remove front cover.
4. Remove gantry display from front cover.
5. Refer to Figure 1. Remove the nine (9) screws holding down the metal cover and the one (1) screw located by the connectors.

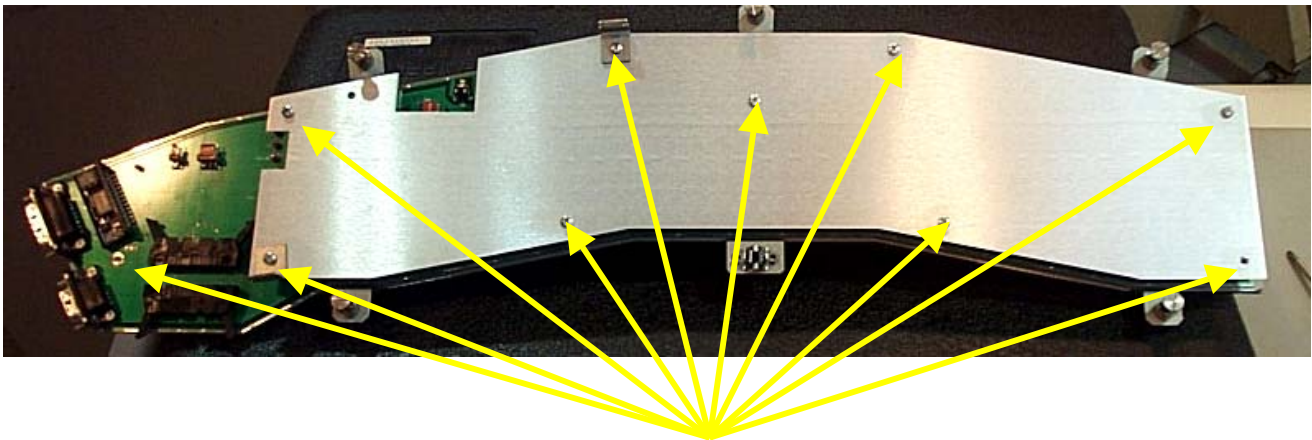


Figure 1. Remove Screws

6. Referring to Figure 2, remove the nine (9) standoffs.

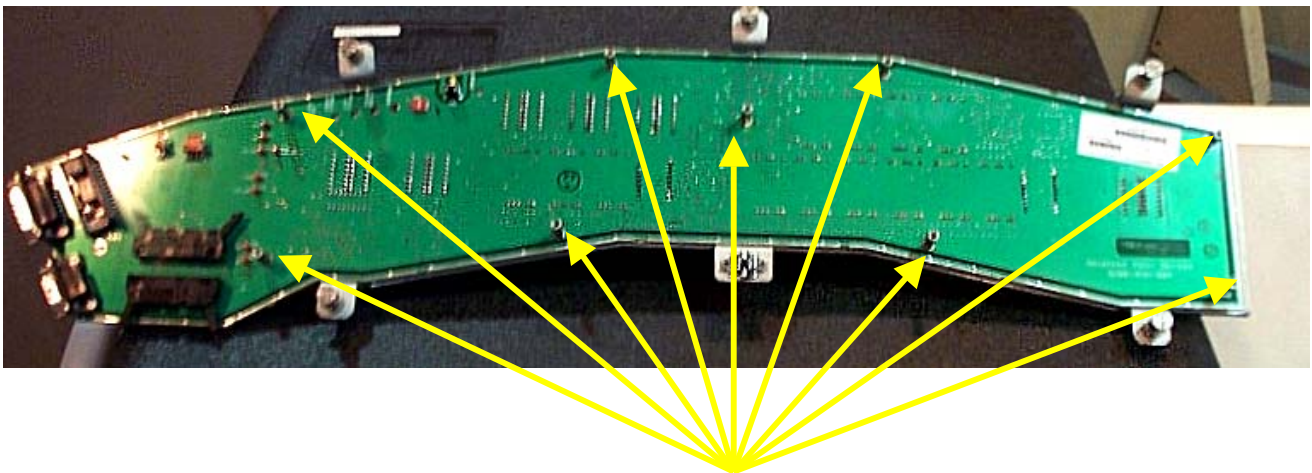


Figure 2. Remove Standoffs

7. Refer to Figure 3. On the front of the display board, locate the screw hole near the connectors as shown. *Carefully* remove the board coating by scraping it away with the blunt tip of a flat blade screwdriver.

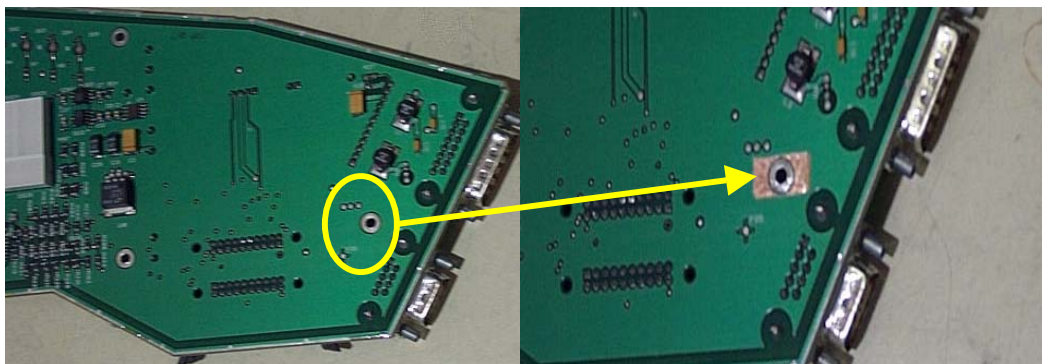


Figure 3. Locate Screw Hole and Remove Board Coating

8. Refer to Figure 3. Cut a piece of copper tape to fit inside of the scraped away area. It should be sized to leave sufficient room for the application of solder.
9. Refer to Figure 5. Apply solder to the copper tape to fasten it to the display board.

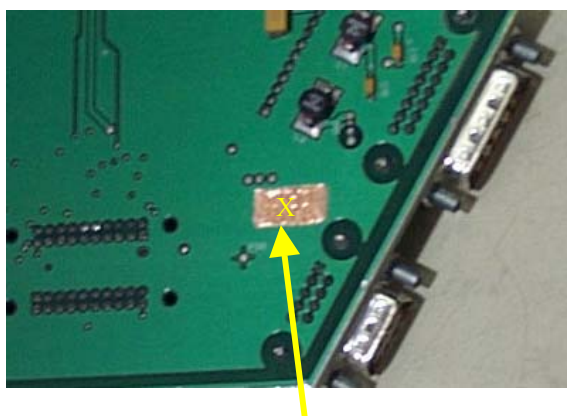


Figure 4. Cut An X

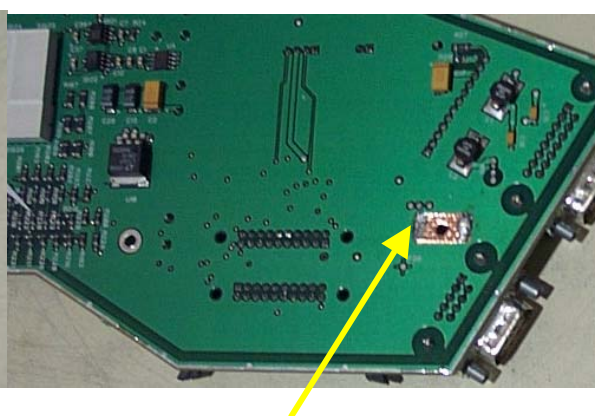


Figure 5. Apply Solder to Fasten Copper Tape to Board

10. Refer to Figure 4. Locate the hole through the tape and, using a sharp Exacto knife, carefully cut an X into the tape over the screw hole.
11. Fold over the copper tape into the hole.
12. Use one of the standoffs to force the tape to conform to the hole's dimensions by forcing it into the screw hole.
13. Add "-2" to the end of the part number imprinted on the board with an indelible ink pen (so it reads: "2256730-2").
14. Reassemble the display by reversing steps 5 and 6.
15. Perform a continuity check from the metal cover to the connector shell to ensure a good electrical connection.
16. Reassemble the display to the gantry cover.
17. Put front cover back on and turn the Axial Drive switch on.
18. Put top and side covers back on.
19. Perform a hardware reset and test scans.



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